Three networks, one question: who do you trust, and how far can you reach?

The internet is the worldwide, public system that connects billions of independently-owned networks so they can exchange data using one shared set of rules.

What this key knowledge covers
This summary distils everything you need for **KK04 — Three networks, one question: who do you trust, and how far can you reach?** in Cyber Security. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Running scenario — SwiftPaws

SwiftPaws is a small Melbourne dog-walking & pet-sitting startup. Customers book walks on a public website. Behind the scenes, an internal staff system stores something genuinely dangerous if leaked: **customers' home addresses, gate codes and house-key locations**. Walkers are out on the street all day and need to pull up the next address from their phones. That one business creates a textbook need for all three networks — we'll follow it the whole way through.

The internet gives you **reach** (anywhere on Earth) but **no trust** (it's public). The intranet gives you **trust** (private, staff-only) but **no reach** (only inside the building). A VPN hands you **both** — internet reach with intranet trust — by digging an *encrypted tunnel* through the public internet.

Lock that one sentence in and most KK04 questions answer themselves.

A global network of networks — public by design

The internet is the worldwide, **public** system that connects billions of independently-owned networks so they can exchange data using one shared set of rules.

Technical underpinnings you must be able to explain:

- **It's a "network of networks."** No single owner. Each organisation runs its own network; **Internet Service Providers (ISPs)** and backbone links join them together so any device can reach any other.

- **Packet switching.** A message is broken into small **packets**. Each packet is routed *independently* and may take a different path; they are reassembled in order at the destination. This is what makes the internet resilient — if one link fails, packets reroute around it.
- **The TCP/IP protocol suite.** Shared rules every device obeys. IP handles *addressing and routing* (which device, which path). TCP handles *reliable delivery* — numbering packets, checking nothing is lost, requesting re-sends, and reassembling them in the right order.
- **IP addresses + routers.** Every device has an IP address. **Routers** read each packet's destination IP and forward it one hop closer, over and over, until it arrives.
- **DNS (Domain Name System).** Translates a human name like swiftpaws.com.au into the numeric IP address machines actually use — the internet's "phone book."

For SwiftPaws, the internet is what lets a customer in Brunswick book a walk, and what lets a walker's phone in Coburg talk to the company's systems at all. It's the reach. The catch is that it offers no built-in privacy.

Lab · Packet switching across the internet

Notice three things: packets take **different paths** (packet switching), they can arrive **out of order** and TCP **re-orders** them, and a **lost packet is re-sent** rather than failing the whole message.

A private internet, walled off for one organisation

An intranet uses the **same technologies as the internet** — TCP/IP, web browsers, web servers, links — but it is **private**: restricted to the authorised members of a single organisation.

- **Same toolkit, closed audience.** Staff open a browser and visit internal sites (rosters, policies, the customer database) exactly as they'd browse the web — but those sites live on *internal* servers and aren't published to the world.
- **Private addressing behind a firewall .** The intranet sits behind a **firewall** that blocks unsolicited traffic from the public internet. Internal devices typically use **private IP addresses** that aren't reachable from outside.
- **Authentication & access control.** You must **log in** with an authorised account; non-staff simply have no way in.
- **Bounded = trusted.** Because membership is controlled and traffic stays inside the organisation, the intranet is treated as a **trusted** zone. This is the opposite trade-off to the internet.

SwiftPaws' intranet is where the dangerous data lives — the **addresses, gate codes and key locations**. Keeping that on a private, firewalled intranet rather than the open web is the first line of defence. The intranet is the trust. Its limitation: by default you can only reach it **from inside** the office network — useless to a walker standing in someone's driveway.

An intranet is **not** "a different internet." It's the *same internet technology* deployed privately for one organisation. Think of it as the internet's rules and tools, used inside a locked building.

The secure tunnel that joins the two

A VPN creates a **private, encrypted connection — a "tunnel" — across the public internet**, so a remote device can reach a private intranet safely, as if it were physically plugged in at the office.

- **Tunnelling (encapsulation).** The VPN *wraps* each of your normal packets inside another packet — like sealing a letter inside an opaque, locked outer envelope before it goes through the public post.
- **Encryption .** The contents of the tunnel are encrypted, so anyone intercepting the traffic on the internet sees only scrambled, meaningless data — not the address or gate code inside.
- **Authentication of endpoints.** Both ends prove who they are before the tunnel forms, so an outsider can't just connect to the intranet by pretending to be staff.
- **Secure remote access.** The result: a SwiftPaws walker on café Wi-Fi or 4G can connect to the company VPN and then reach the internal customer database *as though they were in the office* — across the untrusted internet, but protected.

Virtual — there's no real dedicated cable to the office; the private link is simulated *over* the shared internet. **Private** — encryption + authentication make that simulated link behave like a private, trusted connection. A VPN is how SwiftPaws gets reach and trust at the same time.

Signature lab · The VPN tunnel

Same internet, same café, same eavesdropper. The *only* change is the encrypted VPN tunnel — and that's the difference between leaking a customer's home address & gate code and leaking nothing usable at all.

Compare the three — then sort them yourself

If you can fill this table from memory, you can answer almost any KK04 comparison question.

Sort it: which network does each statement describe?

Drag each card into the matching bin. On a touch screen, tap a card then tap a bin.

One analogy, three roles — tap to flip

The public-mail analogy carries all three concepts at once. Tap each card to reveal how it maps back to the technology.

Anyone can post anything anywhere. Cheap, global, reliable delivery — but a **postcard** can be read by every postie who handles it.

Shared public infrastructure + packet switching. Huge reach, zero built-in privacy — unencrypted data is a readable postcard.

Locked doors, a staff pass at reception. Documents pinned to the wall *inside* can't be seen from the street — but you must be physically **in the building**.

Private, firewalled network with logins. Trusted because it's closed — but only reachable from inside.

Your parcel still drives the **public roads**, but sealed inside a locked, opaque van only the destination can open. Thieves see a van, not the contents.

Tunnelling + encryption over the public internet. You get the road network's reach with a private, trusted ride.

VCE-style questions — with weak vs strong responses

Try each one in your head (or on paper) first. Then reveal a weak answer, a strong answer and the marking guide. Mark your own attempt — your score builds in the dock.

The 60-second recap

Global, public network of networks. Packet switching + TCP/IP. Massive reach, **untrusted** by default.

Same tech, deployed **privately** for one organisation behind a firewall. Trusted, but only reachable from inside.

An **encrypted tunnel** across the internet. Gives remote staff secure access to the intranet — reach *and* trust.

Internet = reach, no trust. Intranet = trust, no reach. VPN = both, via tunnelling + encryption.

Common traps & misconceptions

Exam trap
◆ Security consequence (this is why it's in a *Cyber Security* course) Because internet traffic crosses shared, public infrastructure owned by strangers, it is an **untrusted** environment. Anything sent *unencrypted* can potentially be read or altered by whoever handles the packets along the way. That single fact is the reason intranets and VPNs exist.

Exam trap
⚠ Trap 1 · "An intranet is a smaller internet" Wrong framing. Don't define it by size. Define it by access: a **private** network for one organisation, using the same technologies as the internet but restricted to authorised users.

Exam trap
⚠ Trap 2 · Saying a VPN "hides your identity" and stopping there In this course the examinable point is secure remote access via an encrypted tunnel — it lets remote staff reach a private intranet safely across the public internet. Lead with tunnelling + encryption, not vague "anonymity."

Exam trap
⚠ Trap 3 · Confusing the VPN with the intranet The VPN is **not** the destination — it's the secure path to the destination. Walkers connect *through* the VPN *to* the intranet. The VPN protects the journey; the intranet holds the data.

Exam trap
⚠ Trap 4 · Treating the internet as automatically secure The internet's design goal is **reach and resilience, not privacy**. Always name it as untrusted/public. That's the springboard for justifying a VPN.

Exam trap
⚠ Trap 5 · Listing features with no link to the scenario "Explain" and "justify" questions want the consequence. Don't just say "a VPN encrypts data." Say *why it matters here*: without it, an eavesdropper on public Wi-Fi could capture a customer's home address and gate code.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define · 2 marks

Define the term intranet.

✗ A weak answer

A small version of the internet that a business uses. Defining it by size misses the point and earns little credit.

✓ A strong answer

An intranet is a **private** network that uses the same technologies as the internet (TCP/IP, web browsers and servers) but is **restricted to the authorised members of a single organisation**.

How the marks are awarded

- 1 States it is private / restricted to one organisation's authorised users
- 1 Notes it uses the same technologies as the internet

Q2. Distinguish · 3 marks

Distinguish between the internet and an intranet.

✗ A weak answer

The internet is for everyone and an intranet is for a business. The internet is big and an intranet is small. Two statements sitting side by side isn't a distinction, and 'big vs small' is not an examinable difference.

✓ A strong answer

The internet is a **public** network open to anyone and owned by no single organisation, whereas an intranet is **private** and accessible only to one organisation's authorised members. Consequently the internet is treated as an **untrusted** environment while the intranet, being closed and firewalled, is a **trusted** one.

How the marks are awarded

- 1 Access: public/anyone vs private/authorised members
- 1 Ownership or reach contrast (no single owner / global vs single organisation)
- 1 Trust contrast: untrusted (public) vs trusted (private), stated comparatively

Q3. Explain · 4 marks

A SwiftPaws walker uses café Wi-Fi to look up a customer's address. Explain how a VPN lets them reach the company intranet securely over the internet.

✗ A weak answer

A VPN hides your IP address so hackers can't find you, which keeps you anonymous online. 'Anonymity' isn't the examinable mechanism here. No mention of tunnelling, encryption or the intranet — caps at ~1 mark.

✓ A strong answer

The VPN establishes an **encrypted tunnel** between the walker's phone and the SwiftPaws network: each packet is **encapsulated** and **encrypted** before crossing the public internet, and both ends are **authenticated**. This gives **secure remote access** to the private intranet, so even though the data travels over untrusted café Wi-Fi, anyone intercepting it sees only scrambled data rather than the customer's address.

How the marks are awarded

- 1 Tunnelling / encapsulation of packets
- 1 Encryption of the data in transit
- 1 Provides authenticated, secure remote access to the (private) intranet
- 1 Consequence: intercepted traffic on the public internet is unreadable

Q4. Evaluate · 4 marks

A SwiftPaws manager claims: “We send everything over the internet, so our customers' data is already protected.” Evaluate this claim.

✗ A weak answer

The manager is wrong because the internet is dangerous and full of hackers. A judgement with no supporting reasoning about why, and no protective measure named.

✓ A strong answer

The claim is **incorrect**. The internet provides connectivity and reach but is a **public, untrusted** network with **no built-in privacy** — unencrypted traffic can be intercepted and read in transit. Protection does not come from using the internet itself; it must be added, for example by keeping the data on a **firewalled intranet** and using a **VPN's encryption** (or HTTPS) for remote access. Because the manager assumes the internet is inherently secure, the claim is **overstated and unsafe**.

How the marks are awarded

- 1 Identifies the internet as public/untrusted with no inherent privacy
- 1 Explains unencrypted data can be intercepted/read in transit
- 1 Names a genuine protective measure (VPN encryption / firewall / HTTPS)
- 1 Clear, justified judgement that the claim is incorrect/overstated